GUARDIAN

MAKERS

Installation and user manual

Technical documentation
English edition, October 2026

Operation, wiring and maintenance
Italian / English interface

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GUARDIAN 1.1.0 - TEST4 FIX1
Demonstration technical screens. DEMO/2026 is not a calibration report. Restart preserves PIN and profiles; GUARDIAN reset clears the PIN and invalidates previous profiles. CSV files remain on SD.

Purpose and limits

For maker workspaces

GUARDIAN tracks the environment and displays information from one 3D printer. It helps compare periods and retain readings with their printing context.

Scope

It is an environmental monitor. It is not a reference instrument, a certified safety detector or a replacement for ventilation and exposure assessment.

Printer compatibility

Testing used a Bambu Lab A1. The integration follows Bambu Lab protocols. Other Bambu Lab models require verification; other brands require investigation and possible adaptation.

Evidence in this guide

The photographs show actual prototype readings over partial test periods, not complete 24-hour averages. Network details and credentials are examples. English browser views are labelled separately.

Hardware and assembly

Main components

Freenove ESP32-S3 5-inch display, Sensirion SEN69C, MP1585EN converter, USB-C adapter, rear power socket and microSD card. MP1585EN is the module designation confirmed by the builder.

Enclosure

Keep the sensor low at the rear, on its removable support. The internal partition separates it from the electronics. Keep inlet and outlet paths distinct and unobstructed.

Fasteners

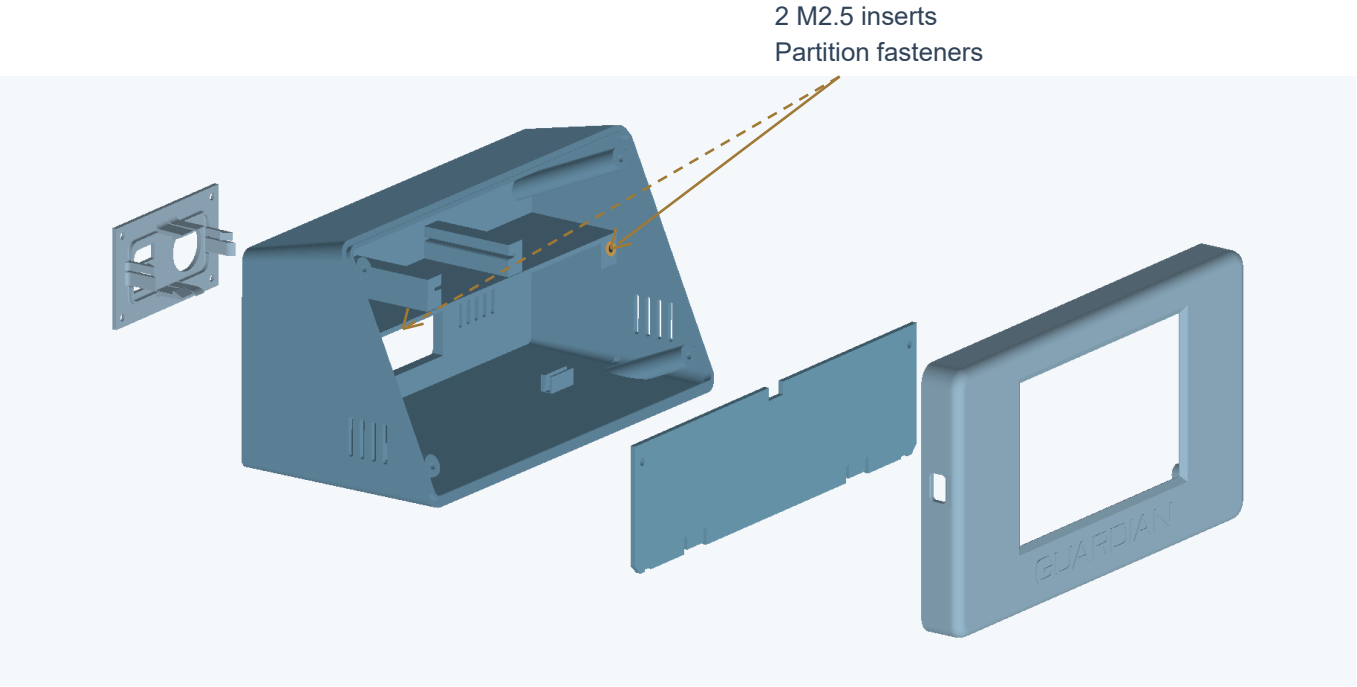
Display and USB board: M2.5 × 5 mm button-head screws. Internal partition and sensor panel: M2.5 × 6 mm countersunk Phillips screws. Case/front panel: M3 × 30 mm socket-head screws.

Inserts and supports

Use the supports and insert seats in the final mechanical files. Do not force screws or overtighten inserts in printed plastic. Verify the assembled clearances before powering the device.

Exploded view of the actual design

Original STL geometry: enclosure, front frame, internal partition and sensor holder. Exploded distances and orientations are illustrative, not assembly dimensions.



Left to right: sensor holder, enclosure, partition, front frame.

Partition: 2 M2.5 heat-set inserts. Both fastening positions are indicated; the dashed arrow points to the position on the opposite side. No inserts are shown in the enclosure screw guides.

Sensor holder: 4 M2.5 heat-set inserts for attachment to the enclosure. Install the inserts from outside before fitting the holder panel, since the closed chamber prevents access from behind.

Enclosure lettering is retained in the original file. Electronics have not been reconstructed using generic components: the actual board is shown on the following pages.

Screws and heat-set inserts

Assembly	Inserts	Screws
Enclosure / front frame	4 × M3 Outer Ø 5 × L 5 mm	4 × M3 × 30 mm Hex socket
Display	4 × M2.5 Outer Ø 3.5 × L 5 mm	4 × M2.5 × 5 mm Button head
Side USB board	2 × M2.5 Outer Ø 3.5 × L 5 mm	2 × M2.5 × 5 mm Button head
Internal partition	2 × M2.5 2 fastening positions	2 × M2.5 × 6 mm Countersunk Phillips
Sensor holder / enclosure	4 × M2.5 Installed from outside	4 × M2.5 × 6 mm Countersunk Phillips

Total: 4 M3 inserts + 12 M2.5 inserts.

Check the outer diameter and length of partition and sensor-holder inserts against the purchased parts and actual seats. An M2.5 thread designation alone does not define the installation bore.

Heat-install before fitting electronics; keep inserts perpendicular and allow cooling. Check remaining wall thickness, screw engagement and clearance. Do not tighten screws against the bottom of blind holes.

Actual board and rear socket

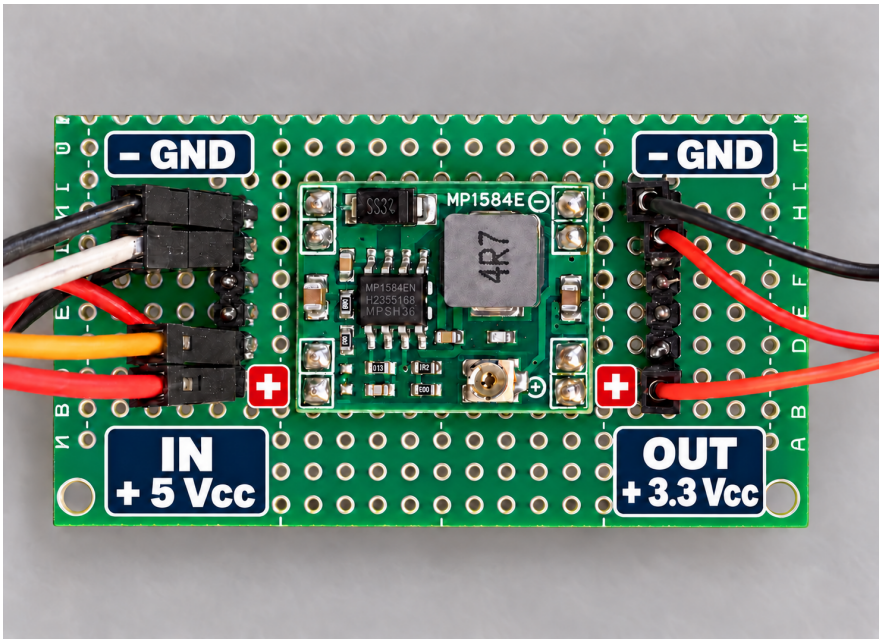


Image of the actual board, with corrected polarity labels. Slide the perfboard into the two rails above the sensor chamber, with IN +5 V on the left and OUT +3.3 V on the right, viewed from inside the open enclosure.

Module identification: the supplied photograph shows MP1584EN. The MP1584 designation, already updated in the wiring diagram, supersedes earlier MP1585EN references. Check the actual component marking before selecting a replacement.

Rear USB-C socket: prewired female panel socket, vendor-rated 5 V / 3 A, 100 mm leads. Fit into the rear opening and check retention. The shown cable uses red for positive and black for GND; verify polarity before connection.

Freenove P5: in the prototype cable, the RED wire is GND. Use only this negative wire for common ground; do not confuse it with the USB socket's red positive lead.

Assembly sequence and final checks

1. Inserts
Install 4 M3 and 12 M2.5 inserts in their seats. Install the sensor-holder inserts from outside.

2. Front frame
Secure the display and side USB board with M2.5 × 5 button-head screws. Do not load the display glass.

3. Power supply
Fit the rear socket and slide the perfboard into its rails, input on the left. Make the parallel Dupont connections on the perfboard: VCC to IN+, grounds to IN−.

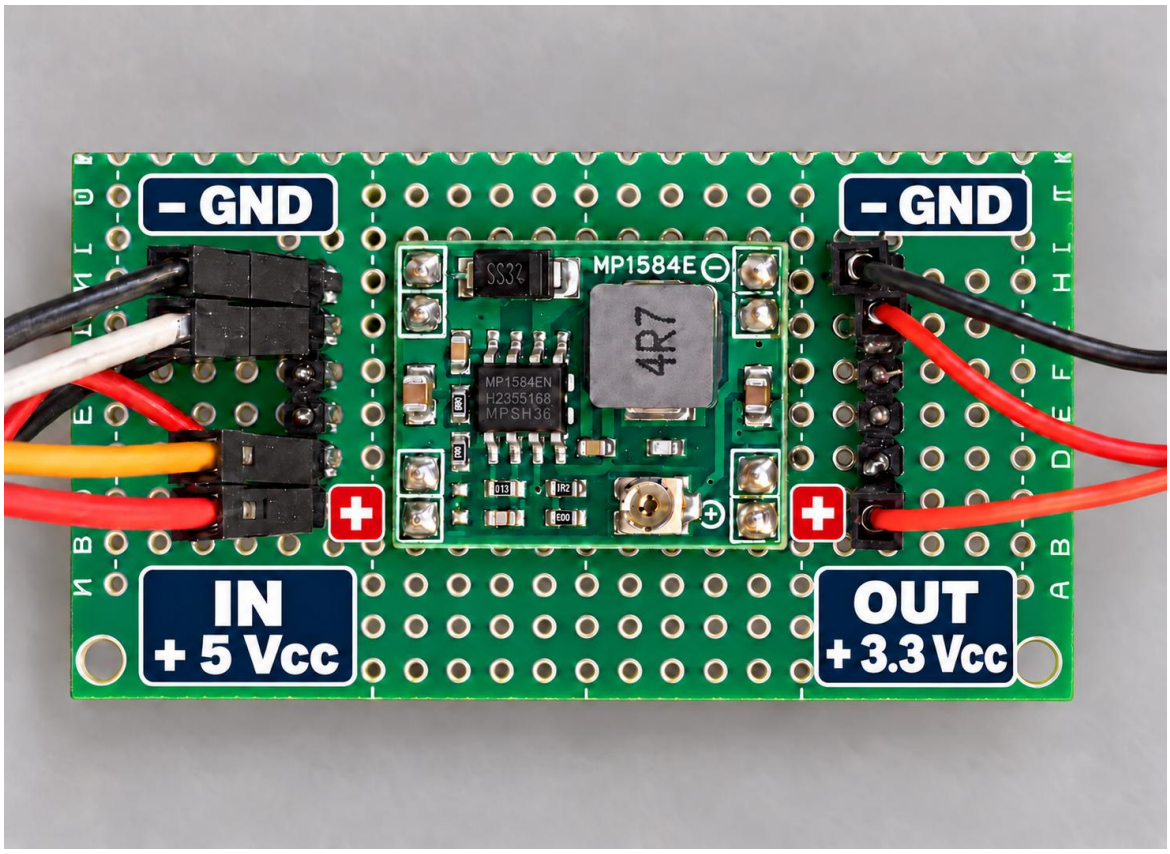
4. Sensor
Verify 3.3 V output before connecting the sensor. Attach its holder to the enclosure with 4 M2.5 × 6 countersunk Phillips screws. Keep air inlet and outlet clear.

5. Partition
Engage the lower tabs, seat the panel and secure it with 2 M2.5 × 6 countersunk Phillips screws in the 2 inserts.

6. Closure
Check for trapped wires. Engage the enclosure lip and close using 4 M3 × 30 hex-socket screws. Fit the TPU cap over the side USB port.

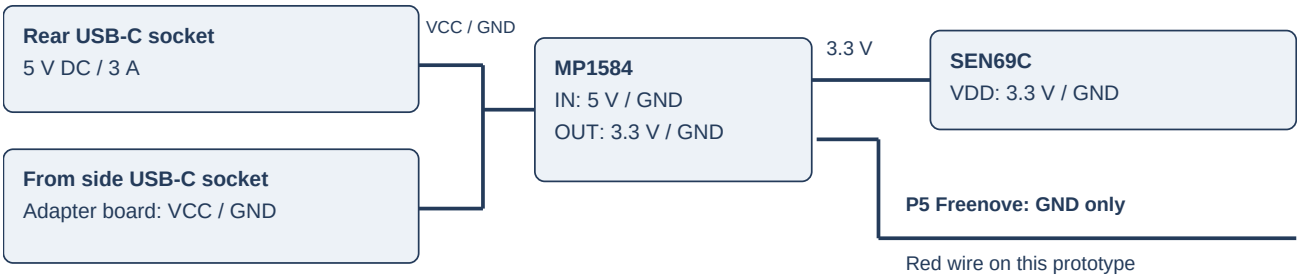
Use only one power source at a time. Both inputs are directly paralleled: disconnect rear power before connecting a PC to the side USB. The cap is not an electrical interlock.

Power supply and wiring



Perfboard 60 x 44 mm | MP1584 22.3 x 16.8 mm | Male Dupont headers

Edited photograph: illustrative labels. Identify terminals from actual board markings.



One power source at a time. Rear and side sockets are directly paralleled, without isolation. Disconnect rear power before connecting the PC to the side socket; disconnect the PC before restoring rear power. The TPU cap identifies the programming port; it is not an electrical interlock.

Perfboard connections. Rear VCC and VCC from the side adapter board connect to IN+; both grounds connect to IN-. OUT+ at 3.3 V supplies SEN69C VDD; OUT- connects to sensor GND and Freenove common ground. Measure and adjust the output before connecting the sensor. Never supply it directly with 5 V.

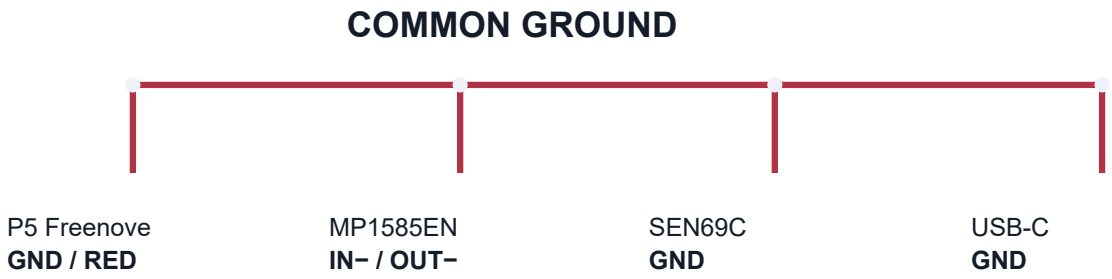
SEN69C signals [S1]. Pin 1: VDD; pin 2: GND; pin 3: SDA to GPIO19 (P4); pin 4: SCL to GPIO20 (P4). Pins 5 and 6 duplicate GND and VDD: follow the manufacturer connector drawing. P5: use GND only; insulate the other three wires. Do not bridge D+ and D-.

Power section update: MP1584, as specified by the builder on 22 October 2026.

P5: common ground only

RED WIRE = GND (NEGATIVE)

Use only the RED GND wire from P5. Leave the other three P5 wires disconnected and individually insulated. The sensor receives power from MP1585EN, not from P5.



Other P5 pins: NOT CONNECTED

P5 3.3 V, IO17 and IO18 are unused in this connection. Follow the board labels, not wire colours. The red/GND identification applies to the cable photographed and confirmed by the builder.

Sensor power

MP1585EN OUT+ adjusted to 3.3 V connects to SEN69C VDD. OUT− connects to sensor GND and the common ground. Check output voltage before connecting the sensor.

Functional ground diagram only. Use one power source at a time; rear and side USB power inputs are directly paralleled.

Adapter and converter installation

P5 ground

Only the RED wire at P5 GND is used for the common ground. P5 3.3 V, IO17 and IO18 remain disconnected and individually insulated. The colour identification applies to the confirmed prototype cable.

Module pin spacing

The MP1585EN module used here has a slightly wider pin pitch than the perfboard grid. Check alignment, use short connections and do not force its pins into misaligned holes.

USB data

Do not bridge D+ and D−. The added jumpers were removed from the prototype adapter to restore communication. Power bridges concern only VCC and GND.

Pull-ups

The Freenove configuration used in this project already includes I2C pull-ups to 3.3 V. Check the actual board before adding parallel resistors. Do not connect I2C pull-ups to 5 V.

One power source

The TPU cap identifies and covers the programming port. It is not an electrical interlock. Remove rear power before attaching a PC, then disconnect the PC before restoring rear power.

Arduino installation

Keep the tested versions

Board package esp32 by Espressif Systems: 3.2.0. Libraries: GFX Library for Arduino 1.6.5; lvgl 9.5.0; TAMC_GT911 1.0.2; Sensirion I2C SEN69C 1.1.0; Sensirion Core 0.7.3; PubSubClient 2.8; ArduinoJson 7.4.3.

Board Manager

Add https://espressif.github.io/arduino-esp32/package_esp32_index.json to Additional Boards Manager URLs, then install esp32 version 3.2.0. [S17]

Library Manager

Select the stated version of each library. Adafruit GFX is not a substitute for GFX Library for Arduino. The IPS display uses GT911 touch. No audio library is required.

Avoid automatic upgrades

Do not use Update All for a working installation. New-version notifications are not faults. Test one change at a time and retain a working copy.

Arduino project and settings

Board options

ESP32S3 Dev Module; OPI PSRAM; Upload Speed 115200; Partition Scheme Huge APP / huge_app. Serial monitor: 115200 baud. Use the remaining Flash/USB options from the working firmware package.

Complete sketch folder

Extract the whole firmware package. Open the .ino file whose name matches the folder. Retain all guardian_* files and bundled resources.

LVGL configuration

Use the working lv_conf.h for LVGL 9.5.0. In the standard arrangement it sits alongside the lvgl library folder. Check the actual include path in compiler output.

Core components

WiFi, NetworkClientSecure, WebServer, ESPmDNS, Preferences, Wire, SPI, FS and SD come with the ESP32 core. FatFs, FreeRTOS and mbedTLS are also included. Do not install competing copies.

Configuration record

Archive the IDE version, every board option and the compile summary. The list here does not replace the complete tested Flash/USB configuration.

Upload and functional checks

1. Back up

Save the existing sketch, lv_conf.h, library versions, settings and CSV files. A normal upload does not format the microSD.

2. Connect

Disconnect rear power, remove the TPU cap and connect a USB data cable to the side port. Select the serial port that appears.

3. Compile

Enable verbose compiler output and use Verify. Check resolved library paths, especially if duplicates exist. Do not upload after a failed compilation.

4. Upload

Close other serial monitors, upload and inspect the startup log at 115200 baud. Confirm the expected firmware version.

5. Check

Test touch, HOME readings, sensor status, SD row growth, CSV download, browser access, lighting and printer data. Retain the previous firmware for rollback.

Updates

Updates currently use USB. OTA updates are not implemented. Preserve flash settings for an ordinary update; RAM history restarts after reboot.

Quick start

1. Position

Place GUARDIAN on a stable surface with the sensor air openings unobstructed. Keep it away from local heat, breathing and direct extraction jets.

2. Power

Use the rear USB-C power input in normal operation. Keep the side programming port covered by the TPU cap. Do not connect the two power sources at the same time.

3. Check measurements

Allow sensor startup and stabilisation. Check HOME and the sensor status. Missing readings do not mean zero.

4. Configure

Set Wi-Fi, time zone, personal thresholds and display lighting. Set up the printer with its local IP address, serial number and LAN access code.

5. Record

Insert and test a compatible microSD card with power off. Confirm that recording rows increase, then download the current CSV from the browser.

Language selection

Select on the display

Open OTHER → SYSTEM, or ALTRO → SISTEMA while in Italian. Choose English or Italiano. The setting is saved without restarting GUARDIAN.

Browser behaviour

PC and smartphone pages follow the display language and reload within approximately three seconds while reachable. Save open forms first: reloading discards unsaved edits.

Persistence

The selected language survives restart. A factory reset returns to Italian. User-entered names, job names and CSV structure are unchanged. Some serial diagnostics remain Italian.

HOME and navigation

Readings

Each tile shows a value, unit and status. An unavailable reading is not a zero measurement.

Navigation

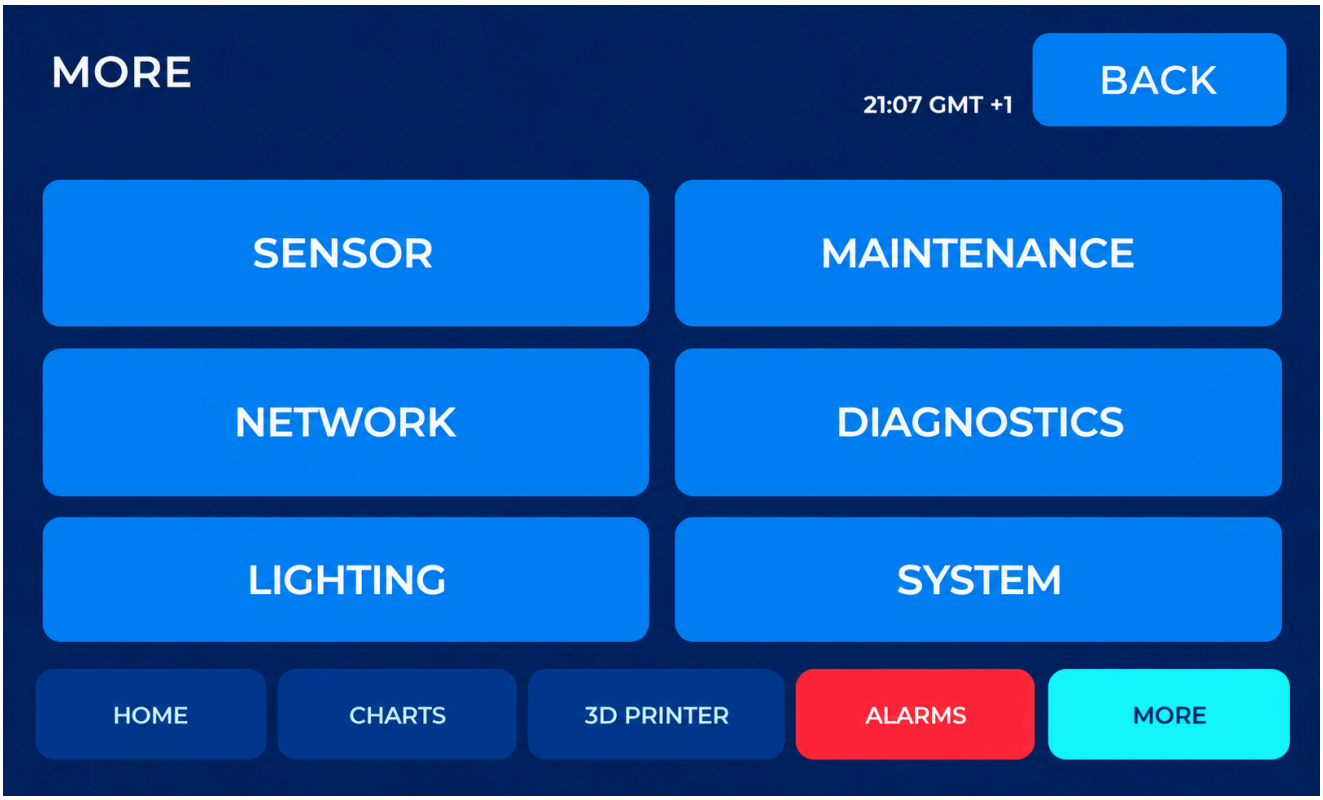
HOME, CHARTS, PRINTER 3D, ALERTS and OTHER provide the main routes. OTHER opens the sensor, network, maintenance, lighting, diagnostics and system pages.

Comfort

Temperature and humidity comfort bands are personal preferences. Outside comfort is distinct from an environmental alert.

Original Italian display, with English instructions. Photo readings cover partial test periods.

The OTHER menu



Select SYSTEM (SISTEMA) to choose the interface language or access restart and reset.

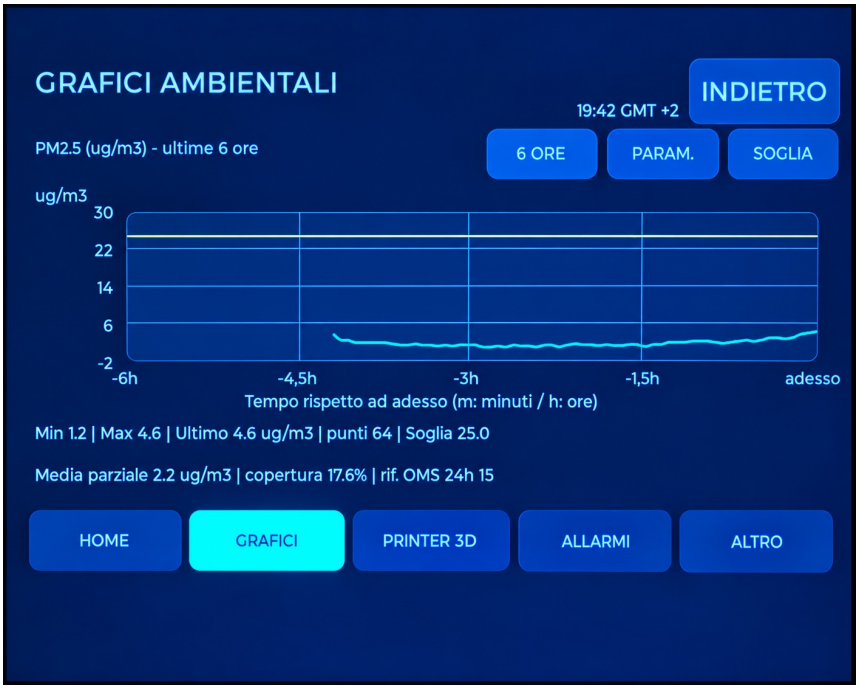
Retouched from the supplied photograph: reflections removed and perspective corrected. Original Italian labels retained.

System and interface language

Select Italiano or English. The choice is saved without restarting. Restart preserves settings; factory reset clears GUARDIAN settings and web access. CSV files remain on microSD.

Retouched from the supplied photograph: reflections removed and perspective corrected. Original Italian labels retained.

Environmental charts



Parameter and period

Choose a parameter and a 2, 6, 12 or 24-hour view. A light grid helps comparison and the vertical axis uses integer labels.

Details below the chart

Current readings and relevant statistics remain available. Integer axis labels do not remove decimals from measurements or CSV data.

History

Chart history resides in RAM and is interrupted by restart. Existing CSV files remain on microSD.

Original Italian display, with English instructions. Photo readings cover partial test periods.

Measurements and averages

Particulate matter

PM1, PM2.5, PM4 and PM10 are optical estimates of mass concentration in µg/m³. WHO 2021 guideline references for PM2.5 and PM10 are 15 and 45 µg/m³ over 24 hours. PM1 and PM4 use personal thresholds. [S1, S6]

A complete 24-hour average

It requires 24 hours of recording with valid data. Before that, the average covers only the available period and is partial. Check missing samples and interruptions. An internal average does not establish regulatory compliance.

HCHO

Formaldehyde is shown in ppb with a 30-minute average. The WHO 2010 reference is 0.1 mg/m³ over 30 minutes, approximately 81 ppb at 25 °C and 1 atm. The conversion depends on temperature and pressure. [S7]

Other channels

CO2 is shown in ppm as a ventilation indicator. VOC and NOx are relative indices, not individual compound concentrations. Temperature and relative humidity use °C and % RH.

Limits

The monitor does not identify all pollutants or directly measure the full ultrafine particle fraction. Low readings alone do not guarantee safe air. [S1, S8, S9]

Thresholds and comfort

Fast adjustment

Use −100, −10, −1, +1, +10 and +100. Holding a button repeats the change; ±1 accelerates while held. Apply/save and check the result.

Browser entry

Enter the numeric threshold directly in the browser. Confirm the save result. A personal operating threshold is not a certified health limit.

Initial settings

PM2.5: 25 µg/m³; CO2: 1000 ppm; HCHO: 81 ppb. Initial comfort bands: 19–26 °C and 40–60% RH.

Persistence and alerts

Comfort preferences require microSD storage. Alert timing and hysteresis reduce oscillation. No alert does not mean that every substance is under control.

Original Italian display, with English instructions. Photo readings cover partial test periods.

PRINTER 3D

Local configuration

Enter the printer IP, serial number, LAN access code and display name. Use the LAN code, not the Bambu account password. One printer at a time; A1 tested and X1 Carbon connectivity verified.

Available information

Progress, layer count, remaining time, temperatures and filament appear when supplied by the printer. The model name does not automatically select a different protocol driver.

Preview and job name

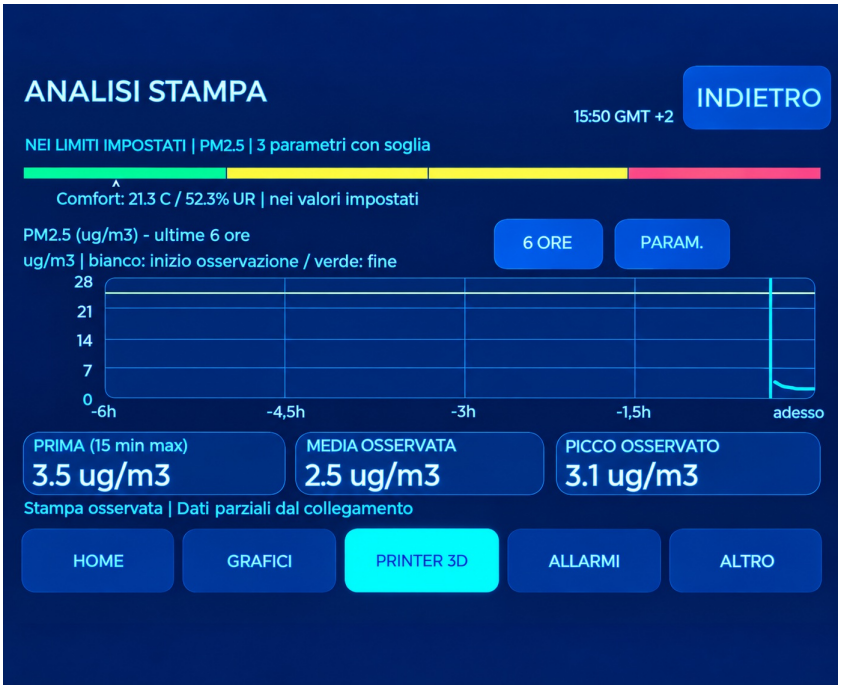
The display preview requires an accessible 3MF. Identical job-name parts separated by “ + ” are displayed once. Different names remain, and the source file is not renamed.

Connection state

Idle or powered-off is distinct from data loss during a print. Old values are identified as stale. Preview retrieval may temporarily suspend and then resume printer data.

Original Italian display, with English instructions. Photo readings cover partial test periods.

Print analysis and environment bar



Session context

Compare the selected environmental parameter with available readings before and during the print, including the session average and peak.

Partial sessions

Starting monitoring after printing has begun, missing samples or restarting makes the comparison partial. Simultaneous variation does not prove the printer caused it.

Summary bar

The bar reflects the worst condition among valid parameters with applicable thresholds. Comfort is separate. It is not a certified health index.

Preserve context

RAM session data is lost on restart. Retain CSV files and printing conditions for later comparisons.

Original Italian display, with English instructions. Photo readings cover partial test periods.

Alert centre

Read the cause

Check the parameter, current value, threshold and data age.

Printer status

A normally idle or powered-off printer is treated differently from loss of communication during printing.

Return to normal

Persistence and a return margin prevent a state change at every small fluctuation.

Informational alerts

GUARDIAN does not stop the printer or replace ventilation and other preventive measures.

Original Italian display, with English instructions. Photo readings cover partial test periods.

Network and browser access

Wi-Fi

Open OTHER → NETWORK. Configure the local network and read the signal strength in dBm, or the disconnected message.

Browser address

Use the address shown by your own GUARDIAN. The credentials and addresses in this guide are examples.

Local access

The local web service uses HTTP. Use a trusted network. For remote access use your network VPN rather than exposing the HTTP port to the Internet.

After changes

Wait for the save result. Connection changes may require a restart; if the IP changes, use the address shown on the display. A guardian-… .local name may not resolve through a VPN.

Original Italian display, with English instructions. Photo readings cover partial test periods.

Browser settings and section order

Display order

Air quality; alert centre; environmental charts; environmental averages; print analysis; 3D printing; SD archive; 3D log download; Wi-Fi connection; Wi-Fi network; printer settings; device access and password; time zone; display lighting; personal comfort; maintenance/calibration; system.

Editing

Thresholds accept direct numeric entry. Available settings include comfort, lighting, time and printer configuration. Check the save result. Device changes synchronise while protecting fields being edited.

Login

Keep your own password private. Remembered access can last up to 30 days depending on session and clock state. Refresh browser cache after a firmware update if an old page remains.

Printer view

Monitoring is read-only for the printer: no movement or stop commands. Preview is on the physical display; GUARDIAN settings remain configurable.

Time zone and display lighting

Time zone

Select a geographic profile and save. Daylight saving follows the selected profile. Synchronisation requires a reachable time source; check the clock after startup.

Manual lighting

Adjust brightness from the display or smartphone browser. Range: 5–100%.

Automatic lighting

Set day/night levels and start/stop times. Defaults: 100% / 15%, night 22:00–07:00. Periods crossing midnight are supported.

Storage

Lighting preferences are on microSD. An invalid clock uses the daytime level. Correct the clock before testing automatic transitions.

Original Italian display, with English instructions. Photo readings cover partial test periods.

Display lighting: validated update

25 kHz backlight PWM

The owner reports that the 5 kHz whine disappeared after increasing the backlight PWM to 25 kHz. Manual brightness and the automatic day/night schedule were tested successfully.

Version identification

This edition describes bilingual GUARDIAN with the 25 kHz modification. The tested sketch is labelled 1.0.0-rc3-pwm25-test1; this label may still appear on the device and in the browser.

Settings and updates

Brightness remains adjustable from the display and browser. This change requires neither factory reset nor SD formatting.

Scope of the test

Functional testing on the owner’s prototype supports keeping this setting. It is not a metrological validation or a certification of the complete product.

microSD reading and test

READ SD

Check capacity, free space, recording status and row count.

TEST SD

Writes, reads, compares and removes a temporary file. Wait for the final result.

Recording

Normally every 30 seconds, with additional events. Interpret errors and dropped records with the serial log.

Removal

Power off before removing the card. Never remove it during writing, testing, downloading or formatting.

Original Italian display, with English instructions. Photo readings cover partial test periods.

CSV download and interpretation

Latest file

Use the dedicated link to the file currently being recorded. Use subsequent archive pages for other CSV files.

Time order

Historical filenames are not necessarily chronological. Use timestamp columns and account for the period before clock synchronisation.

Import

UTF-8 text, semicolon separator and decimal point. Avoid automatic spreadsheet conversions that change numbers or dates.

Fields

Logs include time, uptime, events, validity, environmental values with units, sensor status and alerts. Extended records include print context, partial-session status, job, filament and progress when available.

Retention

An empty field is not zero or proof of no emissions. Back up periodically. Charts do not replace the archive; formatting removes files and SD preferences.

Formatting the microSD

Before formatting

Back up all files and note lighting/comfort settings. FORMAT erases the entire card contents, not just CSV files. Keep power stable.

Request and confirmation

Open the SD page, choose FORMAT and read the confirmation. A request message is not the final result.

Wait

Do not switch off, remove the card or start other SD operations. Wait for formatting and verification to finish.

Verify success

Run READ SD and TEST SD, check that recording resumes and download the new CSV.

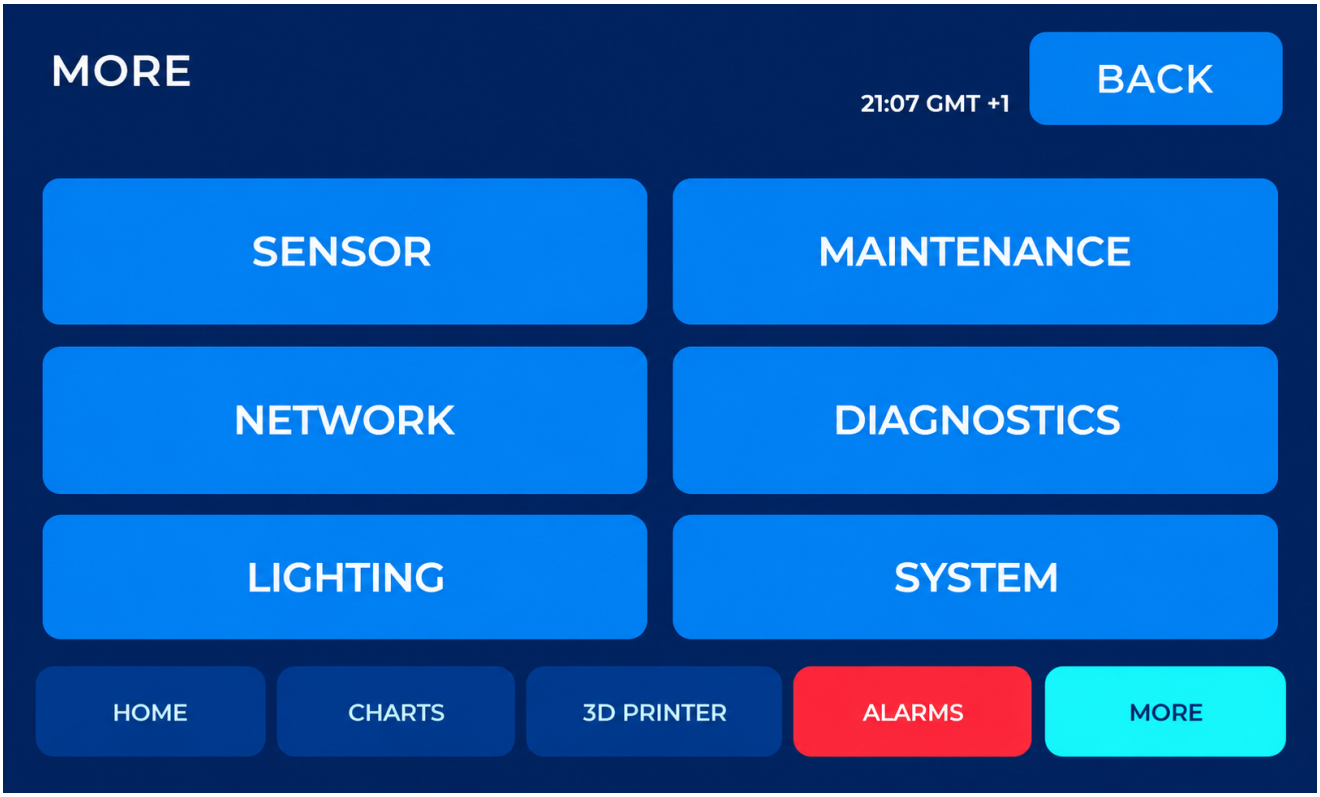
If it fails

Retain the display message and GUARDIAN SD FORMAT serial lines. FatFs 3 alone does not prove card failure. Check power and access before retrying.

Effects

SD preference files are erased. Values can remain in RAM until restart; save them again if required. This is separate from GUARDIAN reset and CO2 reset.

A clear technical workflow



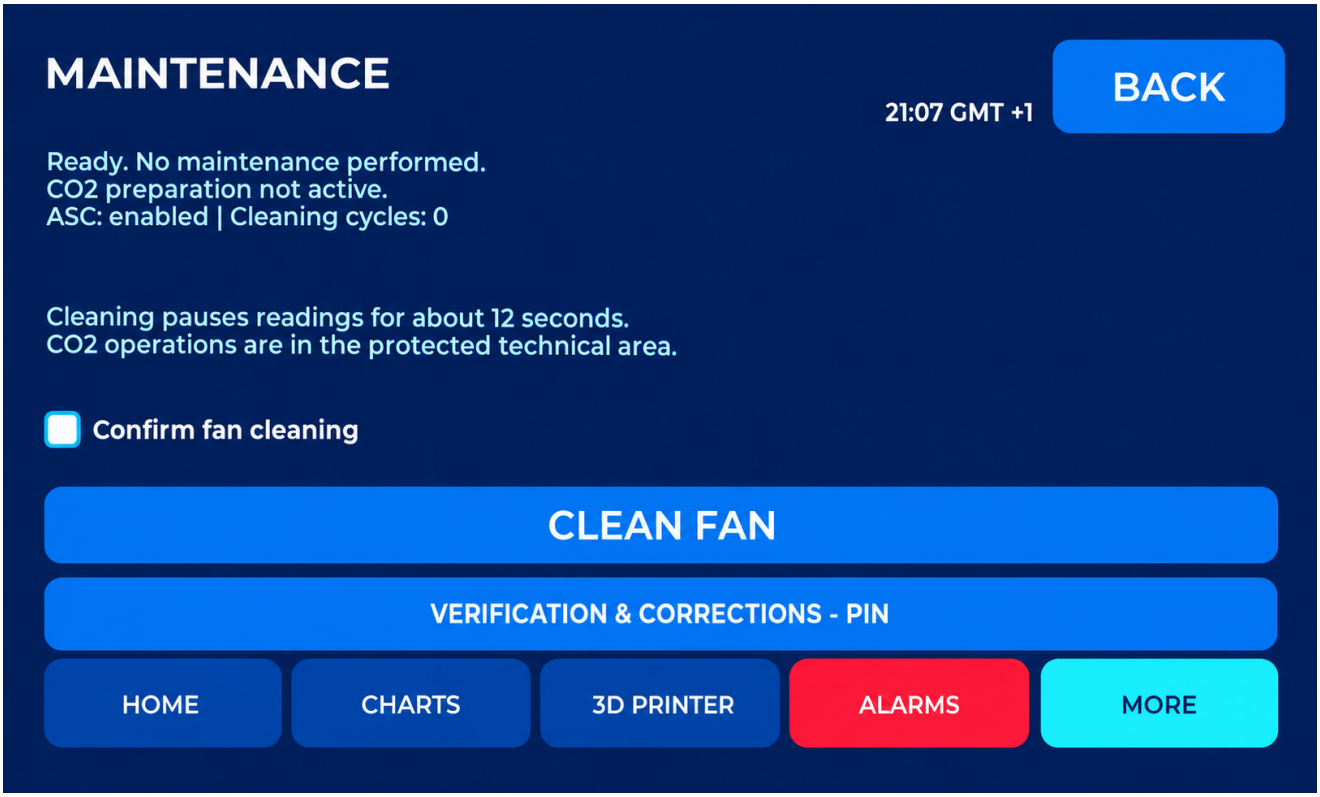
Cleaned demonstration screen based on a real photograph • DEMO/2026

Open MORE → MAINTENANCE → VERIFY & CORRECT. Measurement settings are grouped in a protected area, separate from everyday viewing.

First check SEN69C presence and status on the SENSOR page. Inspect wiring only with the power disconnected.

This edition describes GUARDIAN 1.1.0 and the functions tested in build TEST4 FIX1. Readings and DEMO/2026 report identifiers are examples, not laboratory results.

Maintenance and preliminary checks



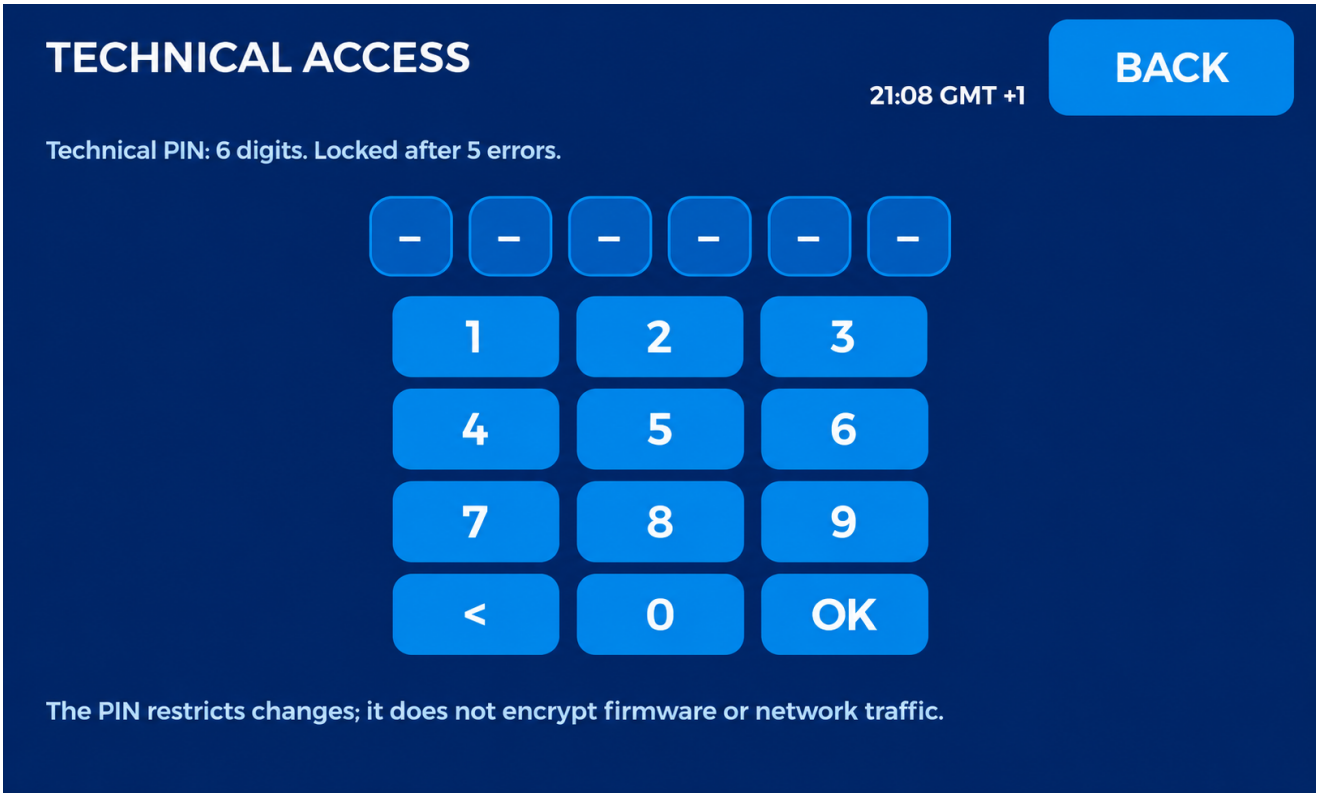
Cleaned demonstration screen based on a real photograph • DEMO/2026

Fan cleaning is a sensor maintenance operation. Start the cycle and wait for measurements to resume. Completion does not certify cleaning effectiveness.

Before comparing instruments, allow stabilisation and record placement, environmental conditions, sampling interval and reference instrument.

Compare the same quantity and unit. For particulate matter, aerosol composition also affects the comparison. Do not compensate for an assembly fault with a correction coefficient.

Technical access: six-digit PIN



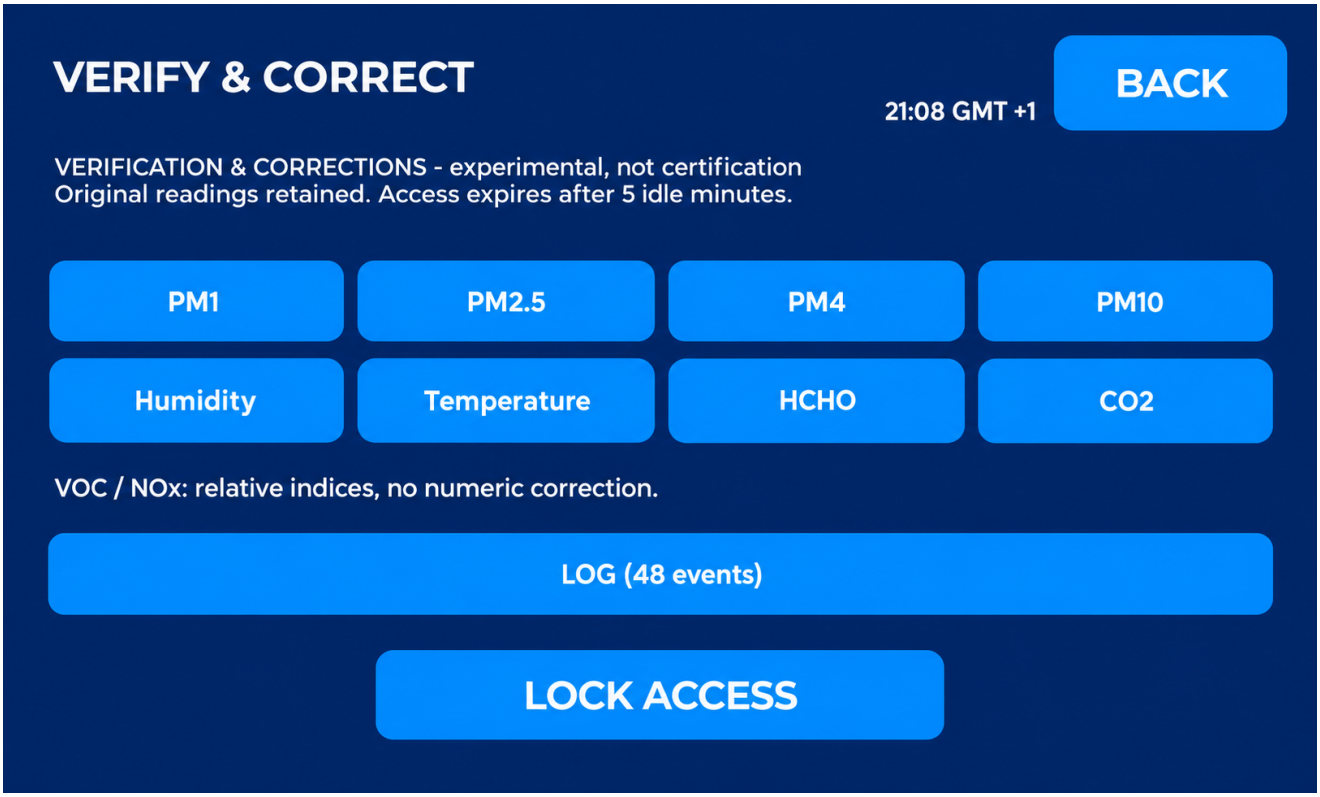
Cleaned demonstration screen based on a real photograph • DEMO/2026

On first access, choose a six-digit PIN and enter it twice. The technical PIN is separate from network credentials and web access.

After five incorrect attempts, wait 60 seconds before retrying. Technical access expires after five minutes of inactivity. Keep the PIN safely.

The PIN restricts interface changes; it does not encrypt memory or enable Secure Boot. Do not change corrections merely to obtain more reassuring readings.

Which parameters can be corrected



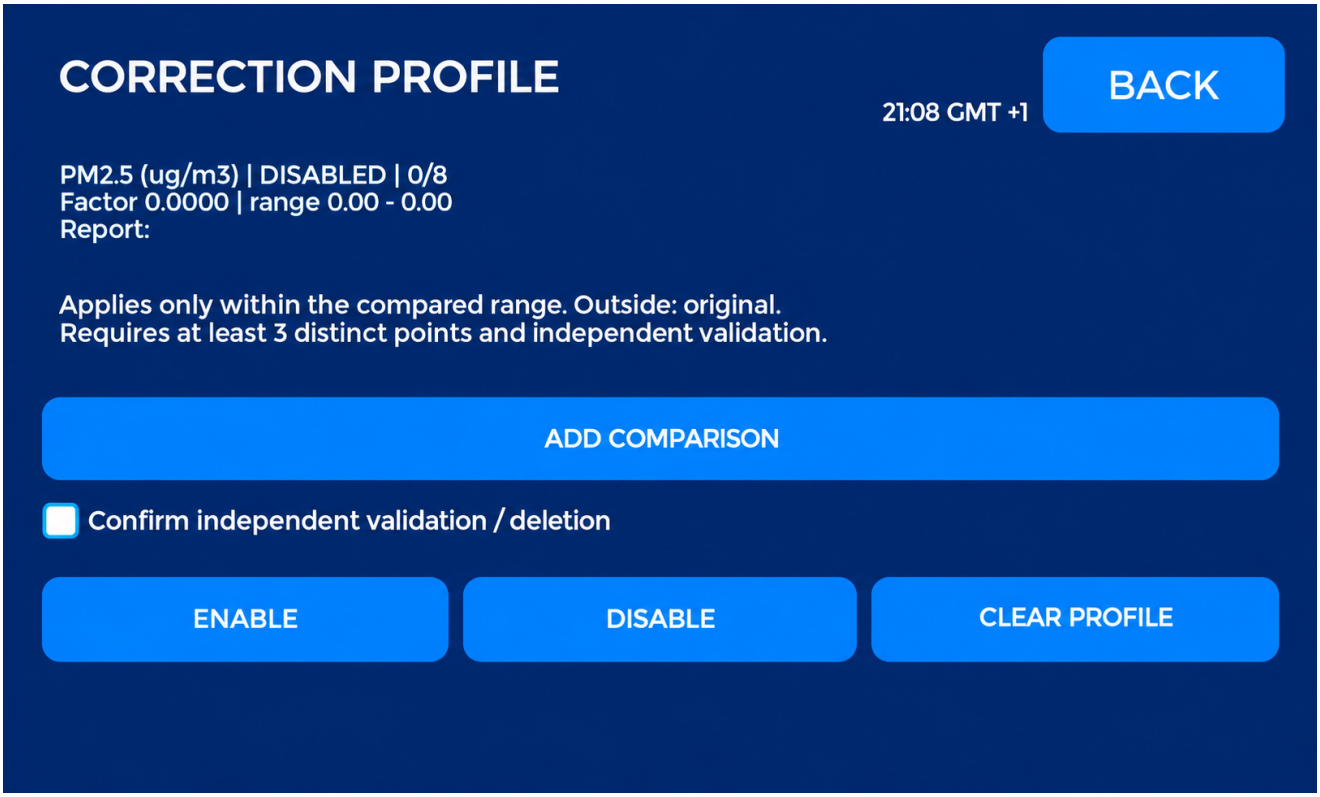
Cleaned demonstration screen based on a real photograph • DEMO/2026

PM1, PM2.5, PM4, PM10, temperature, relative humidity, HCHO and CO2 are available. VOC and NOx are relative indices and have no user correction in this function.

Each parameter has a profile containing comparisons, factor or offset, valid range and report reference. Up to eight comparisons can be retained per parameter.

Particulate channels use a multiplicative factor; the other channels use an additive offset. The function documents a comparison; it does not itself certify the product.

Profile: enabling and valid range



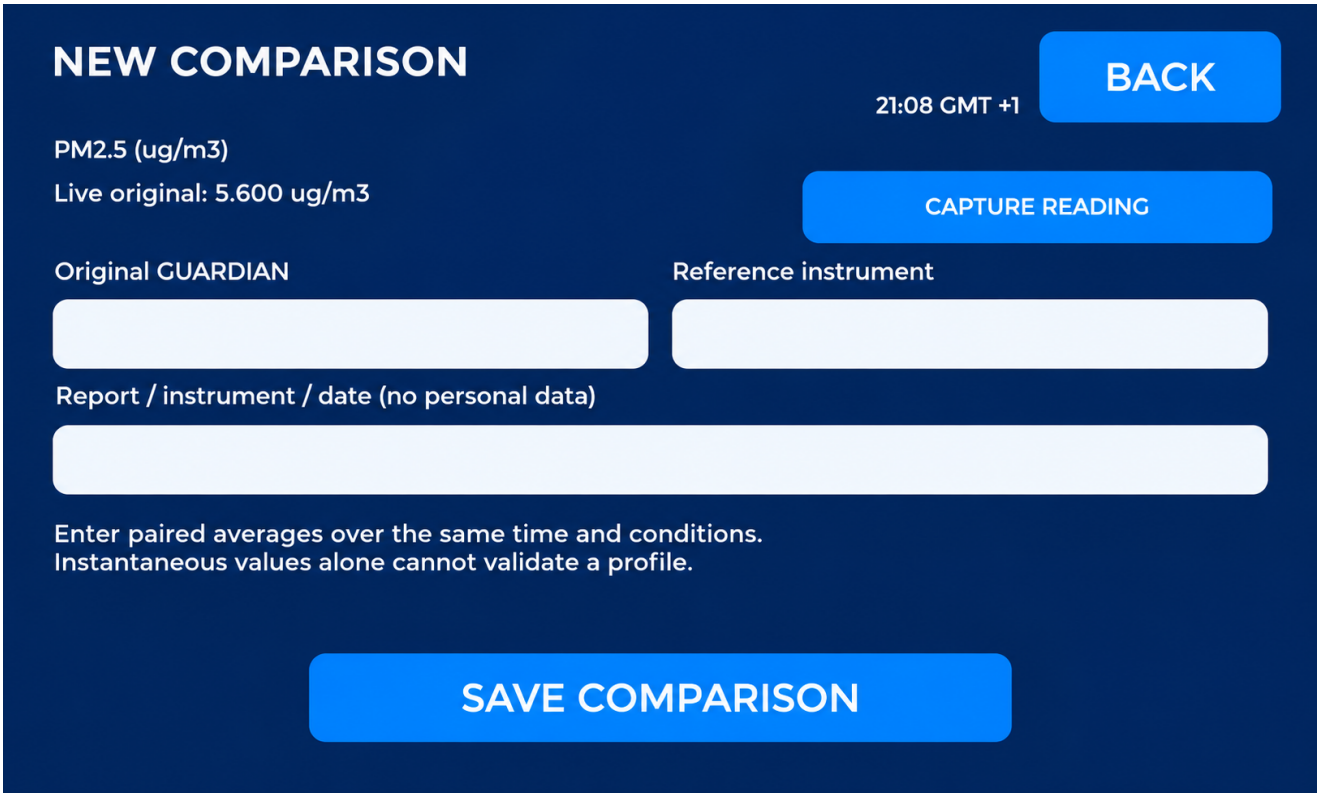
Cleaned demonstration screen based on a real photograph • DEMO/2026

Enabling a correction requires at least three distinct original readings, a report identifier and confirmation of independent verification. These are software requirements, not a complete metrological procedure.

Correction applies only within the range of the compared original readings. Outside that range, the original reading is returned.

DISABLE retains comparisons but suspends correction. CLEAR PROFILE deletes points and the coefficient after confirmation. An asterisk indicates an enabled profile, not proof that the current reading is within its valid range.

New comparison: original and reference



Cleaned demonstration screen based on a real photograph • DEMO/2026

GUARDIAN ORIGINAL shows the current reading before GUARDIAN software correction. It is not a first-start reading and does not undo internal SEN69C calibration.

Capture the reading into the comparison field, then enter the reference instrument reading in the same unit. Prefer averages over the same interval, with stable conditions and comparable placement.

The report field accepts /, for example DEMO/2026. For an actual verification, enter the real report reference. Save and check that the comparison count has increased.

CO2: two different operations

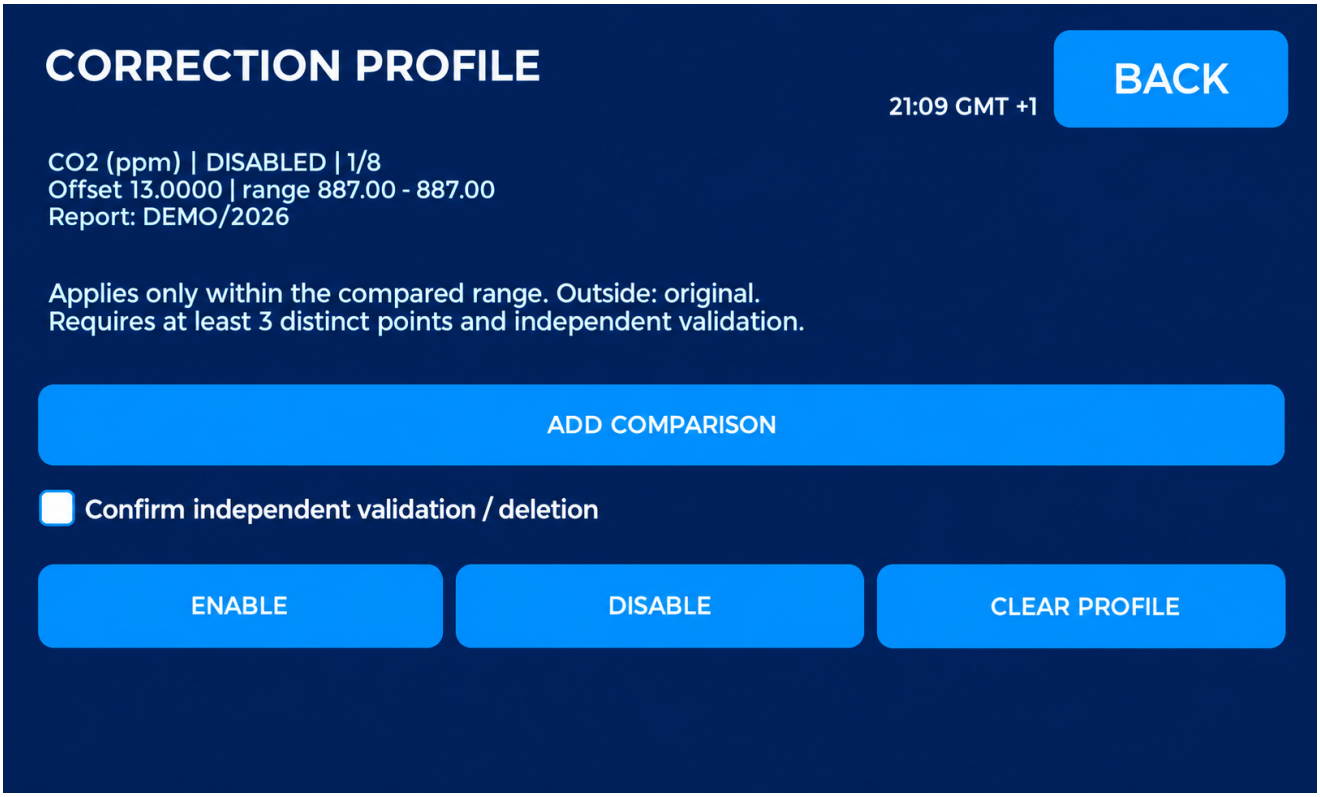
Cleaned demonstration screen based on a real photograph • DEMO/2026

The CO2 entry groups the software correction profile and internal sensor operations. Choose the path appropriate to the intended verification.

Software correction adds an offset within the verified range and can be disabled. Internal calibration instead changes the sensor’s CO2 channel.

Do not stack the two operations: disable the software CO2 correction before internal calibration. Always record which operation was performed and the reference used.

CO2 example: one point is not enough



Cleaned demonstration screen based on a real photograph • DEMO/2026

The screen shows one comparison: an original reading of 887 ppm and a calculated +13 ppm offset. The profile is DISABLED and its range is limited to that single point.

The software enabling requirements are not yet met. At least three distinct original points, a report and independent verification are needed; repeating the same reading does not extend the range.

Select points representative of intended use within a suitable comparison plan. A laboratory can define points, repeats, uncertainty and acceptance criteria.

CO2 comparison and data retention

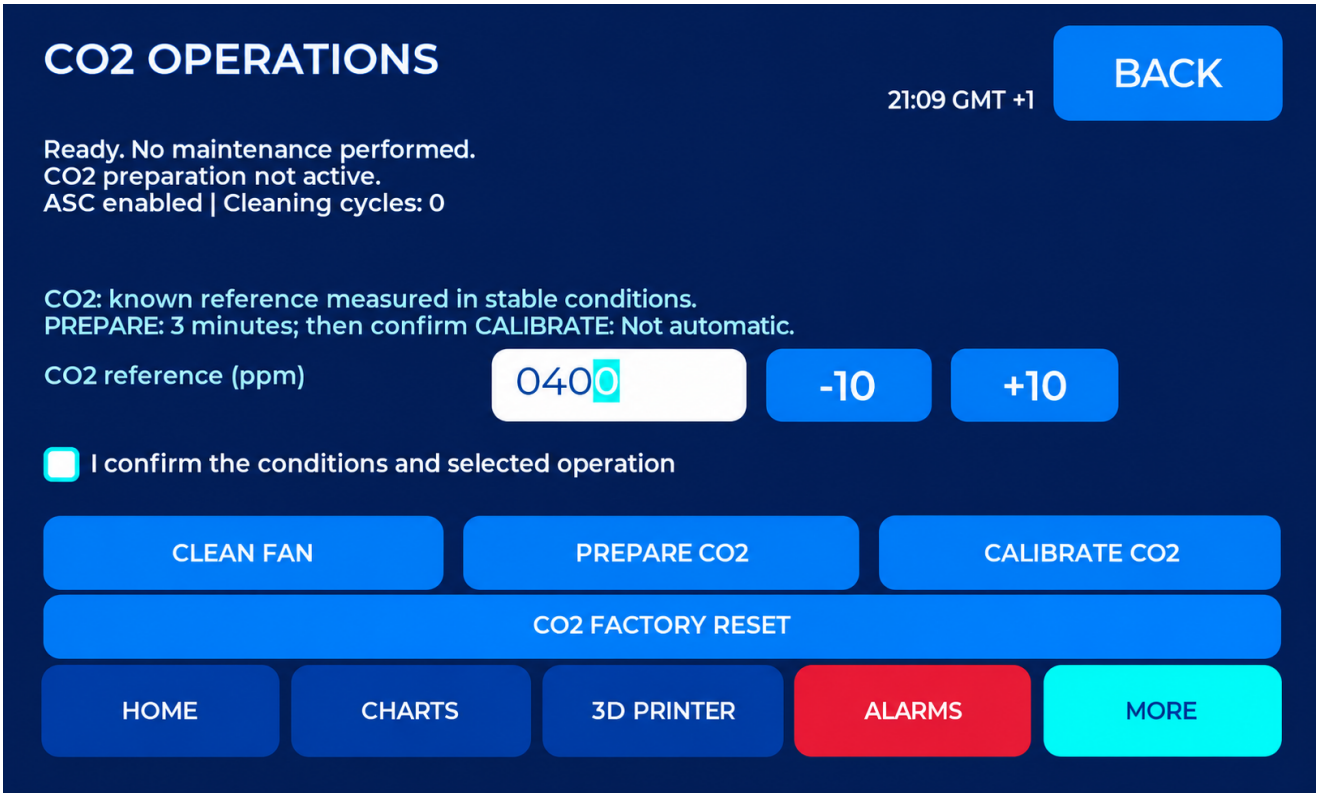
Cleaned demonstration screen based on a real photograph • DEMO/2026

The original reading continues updating; the captured comparison value remains the selected value. Check the reference before saving.

Profiles and the technical log are stored on microSD. A present, working card is required to retain changes; wait for the save result.

Restarting preserves the PIN and profiles. Export data before formatting: formatting is separate from GUARDIAN reset and deletes card files. Recheck active profiles after any intervention.

CO2 operations: prepare and verify



Cleaned demonstration screen based on a real photograph • DEMO/2026

Use a reliable, known reference within the allowed 400–2000 ppm range. Outdoor air does not automatically mean 400 ppm. Avoid breath and local sources and wait for stable conditions.

PREPARE starts a three-minute wait. READY alone does not guarantee equilibrium: verify conditions, confirm and start CALIBRATE. Wait for the result and resumed measurements.

CO2 RESET concerns the sensor channel, not the whole GUARDIAN. After success, the software CO2 profile is also cleared. Allow stabilisation and repeat the comparison. The command does not issue a certificate.

Technical log and traceability

Technical log: new optimized photo (Italian display).

The log retains the last 48 events: comparison additions, profile enabling, disabling and clearing.

To export: open the browser on the same network, MAINTENANCE > Verification & corrections (PIN). Enter the technical PIN and select UNLOCK. Scroll to Change log and select DOWNLOAD CSV. GUARDIAN_technical_audit.csv is saved in Downloads; exporting does not clear the log.

Columns show sequence, UTC, uptime, parameter, action, origin, before/after and report. Before/after means factor or offset. Keep the CSV with the laboratory report: it is not an immutable archive or a certificate. The new photo is optimized for readability; the display is in Italian.

Diagnostics

DISPLAY

Cycle the colours and check uniformity. Distinguish momentary transition artefacts from persistent lines. Document any persistent defect.

TOUCH

Touch different areas and check position/response. CLEAR clears the test view; it does not reset GUARDIAN.

SD

READ, TEST and FORMAT have different effects. Wait for each final result.

Serial log

At 115200 baud inspect uptime, memory, sensor samples, errors and network events. A restarted counter indicates a new boot; printer data loss alone does not prove GUARDIAN rebooted.

Original Italian display, with English instructions. Photo readings cover partial test periods.

Restart, reset and loan preparation

Restart

Restarts GUARDIAN, interrupting RAM history and the active session. Previously written files remain.

Factory reset

After confirmation, clears network, printer, thresholds and preferences, removes the technical PIN and invalidates previous correction profiles. CSV files remain on microSD.

Data on microSD

Factory reset does not automatically erase CSV files or perform CO2 reset. Back up and separately format the card, or use a new card when lending the unit.

Before lending

Use firmware without embedded personal credentials, reset GUARDIAN and verify a fresh configuration is required. Inspect remaining files and job names. Reset/format are logical cleanup, not forensic erasure.

Unresponsive interface

An on-screen reset is unavailable if the interface is frozen. Record the fault and restart physically if necessary.

Troubleshooting

Serial port missing or busy

Try a direct data cable and another USB port. Remove data-line jumpers from the adapter. Close other serial monitors and select the port that appears now.

Display or sensor absent

Check OPI PSRAM, library versions and boot log. With power off, verify sensor 3.3 V, common GND and SDA19/SCL20.

TLS or stale printer data

Check printer power, local IP, network and LAN access code. Bambu Studio via cloud can work even when the local connection fails.

No preview

Retain the log and check the active job and 3MF availability. Missing preview does not mean the print stopped.

CSV or slow listing

Use the current-file link and archive page controls. Check card accessibility and status.

Browser or language outdated

Check the device address, reconnect and refresh cache. Verify the language on SYSTEM. Save edits before changing language.

Measurement quality and documentation

After servicing

Test startup, touch, sensors, SD/CSV, browser, lighting and printer data. Check a new file and increasing row count after SD work.

Keep a configuration record

Archive the IDE, board options, lv_conf.h, library versions and compile summary. Record physical wiring changes.

Validation

Compare with suitable references under documented conditions. Characterise warm-up, enclosure heating and airflow. Software testing does not establish metrological accuracy.

Calibration document

The attached Sensirion statement concerns factory calibration of the listed sensor families. It is not an individual calibration report for the complete GUARDIAN device. [S2]

Product conformity

Component REACH/RoHS documentation does not certify the assembled product. RED, RoHS, WEEE and REACH are references for an eventual finished-product assessment; no completed GUARDIAN certification is claimed. [S11–S16]

Individually tested by Sensirion

Factory calibration and individual testing

Sensirion states that each listed device is individually tested after calibration. SEN69C is included in the SEN5x/SEN6x declaration. [S2]

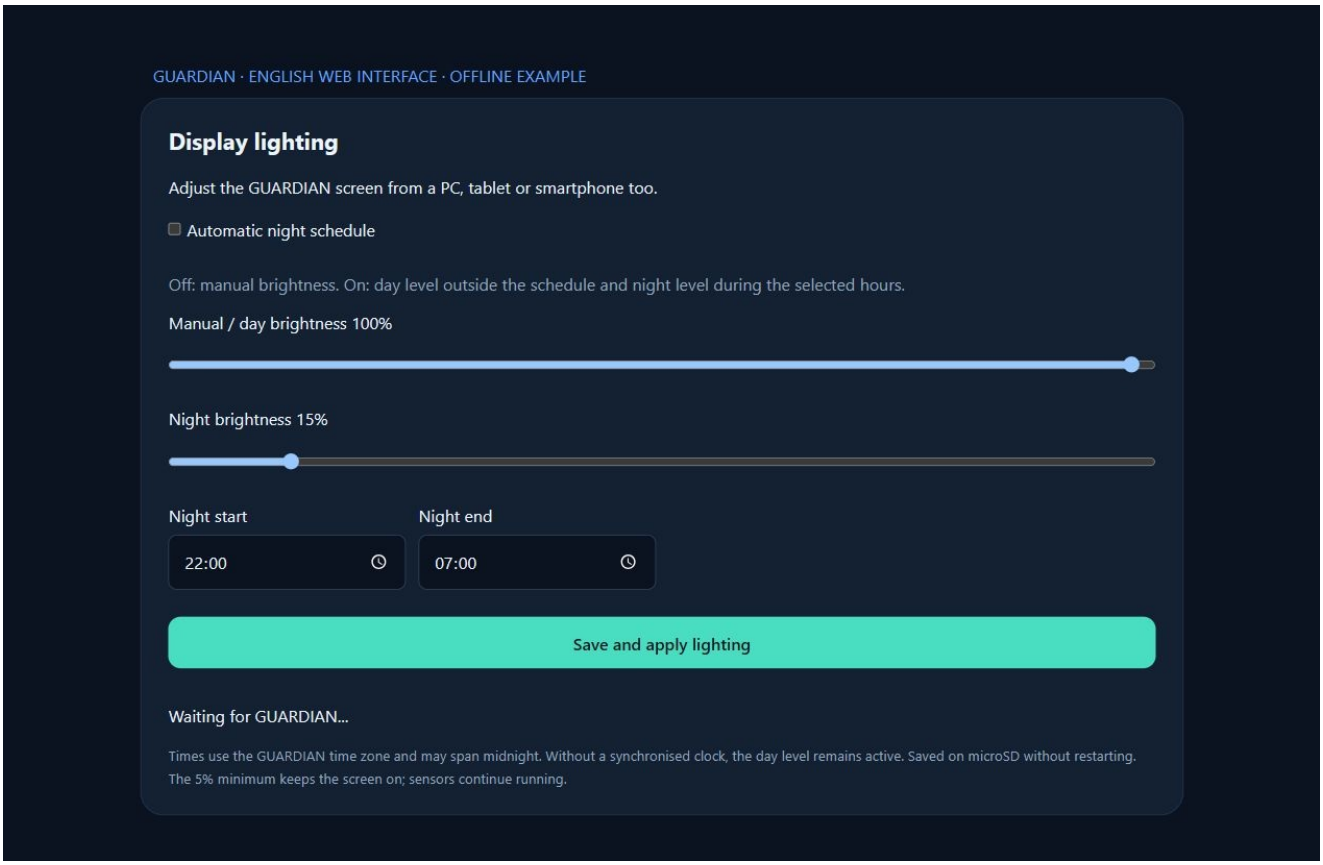
Scope of the statement

The May 2026 declaration is titled Particulate Matter Sensors. It must not be presented as certification of every gas channel or of the complete GUARDIAN device.

Manufacturer evidence

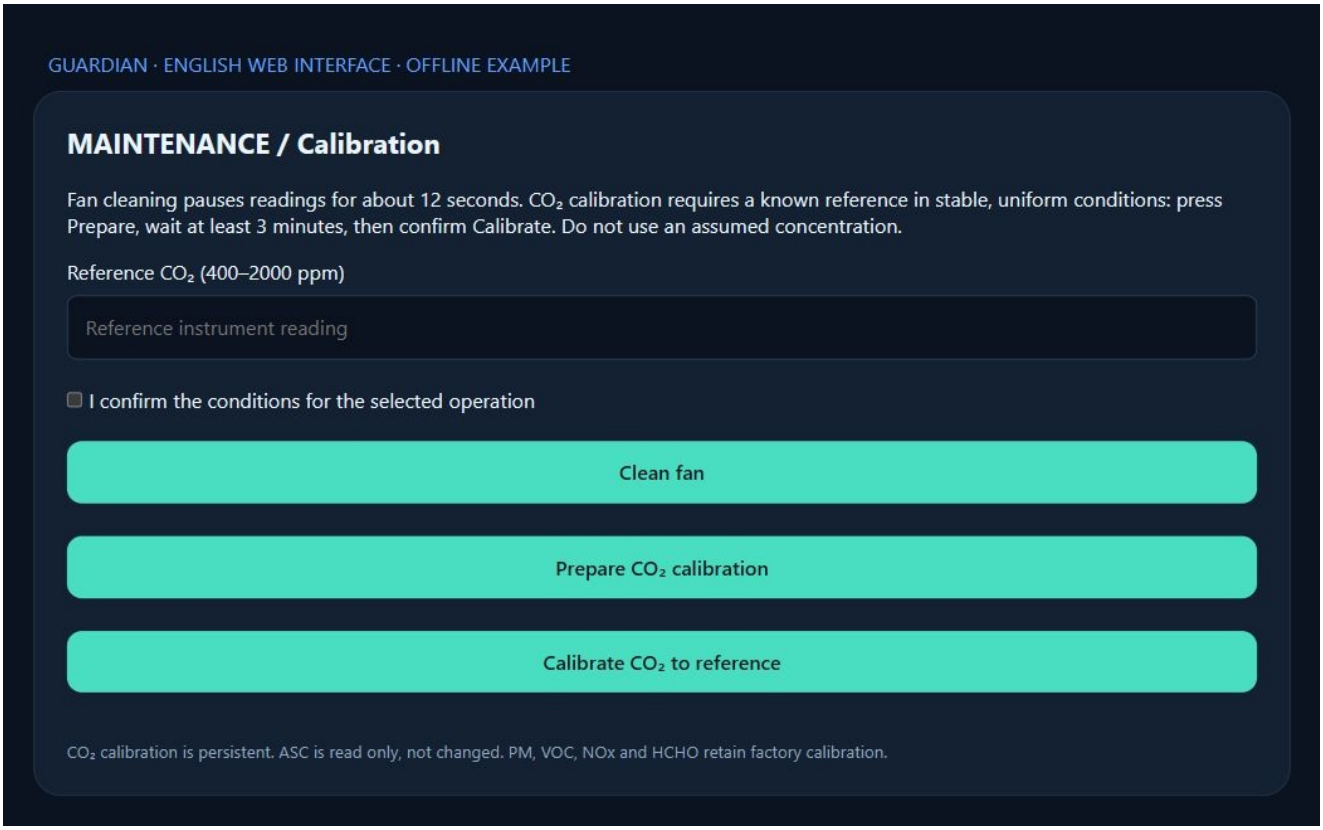
The original statement is attached to the manual. Enclosure integration, airflow and operating conditions still require their own checks.

English interface: display lighting



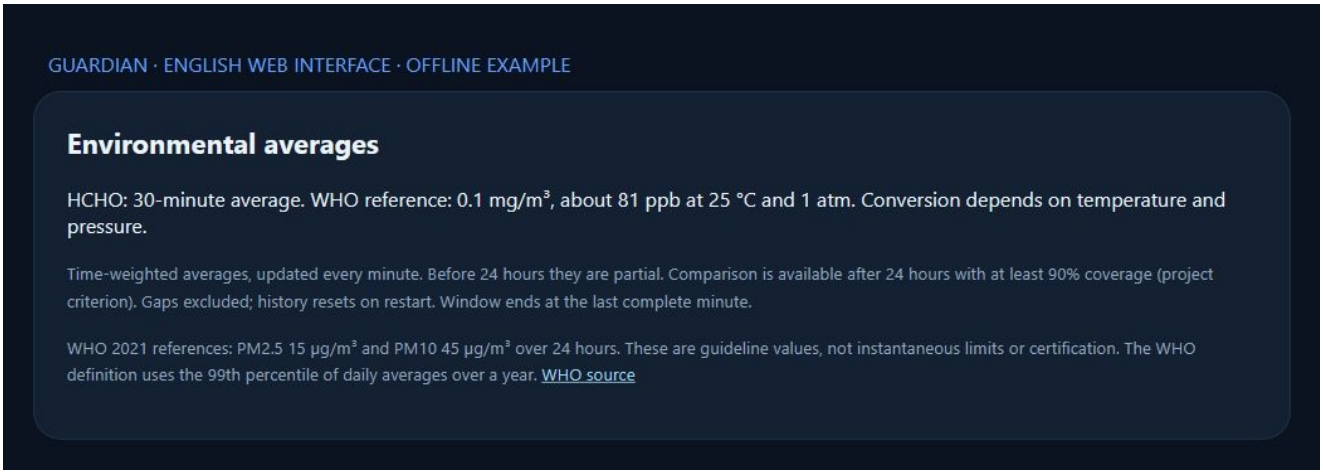
Offline example rendered from the English firmware web page. No live readings or personal credentials. The layout adapts to browser width.

English interface: maintenance



Offline example rendered from the English firmware web page. No live readings or personal credentials. The layout adapts to browser width.

English interface: averages



Offline example rendered from the English firmware web page. No live readings or personal credentials. The layout adapts to browser width.

Sources and reference documents

S1 - Sensirion SEN6x datasheet
Open original source

S2 - Sensirion factory calibration statement, SEN5x/SEN6x
Open original source

S3 - Sensirion SEN6x mechanical design and assembly
Open original source

S4 - Sensirion sensor specification statement
Open original source

S5 - Sensirion temperature compensation
Open original source

S6 - WHO Global Air Quality Guidelines, 2021
Open original source

S7 - WHO indoor air quality: selected pollutants, 2010
Open original source

S8 - EPA research on 3D-printing emissions
Open original source

Manufacturer and regulatory references do not certify the complete GUARDIAN device. Original documents retain their source language.

Sources and reference documents

S9 - EPA Air Sensor Toolbox
Open original source

S10 - UBA: CO2 and ventilation
Open original source

S11 - Radio Equipment Directive 2014/53/EU
Open original source

S12 - RoHS Directive 2011/65/EU
Open original source

S13 - Delegated Directive (EU) 2015/863
Open original source

S14 - WEEE Directive 2012/19/EU
Open original source

S15 - REACH Regulation (EC) 1907/2006
Open original source

S16 - Sensirion REACH/RoHS declaration
Open original source

Manufacturer and regulatory references do not certify the complete GUARDIAN device. Original documents retain their source language.

Sources and reference documents

S17 - Espressif Arduino ESP32 installation
Open original source

S18 - Freenove FNK0115 repository
Open original source

S19 - Sensirion SEN69C product information
Open original source

Manufacturer and regulatory references do not certify the complete GUARDIAN device. Original documents retain their source language.

Sensirion statement

Calibration Certification – Particulate Matter Sensors

SENSIRION
THE SENSOR COMPANY

Calibration Certification

Name and address of the manufacturer: Sensirion AG
Laubisruetistrasse 50
CH-8712 Switzerland

Description: Particulate Matter Sensors

- SEN50, SEN54, SEN55
- SEN62, SEN63C, SEN65, SEN66, SEN68, SEN69C

The above mentioned products are calibrated to meet the specifications according to the corresponding Sensirion data sheet. Each device is individually tested after its calibration.

Sensirion uses transfer standards for the calibration. These transfer standards are themselves subject to a scheduled calibration procedure. The calibration of the reference itself used for the calibration of the transfer standards is performed by an ISO-9001:2015 registered calibration system.

The accreditation body is full member of the International Laboratory Accreditation Cooperation (www.ilac.org). Calibration certificates issued by facilities accredited by a signatory to the ILAC Mutual Recognition Arrangement (MRA) are accepted by all signatories to the ILAC MRA.

This provides traceability of measurement to recognized national standards and to units of measurement realized at the "National Physical Laboratory" (NPL) or other recognized national standards laboratories like "Physikalisch-Technische Bundesanstalt" (PTB) or "National Institute of Standards and Technology" (NIST).

Staefa, May 2026

Stephan Weber,
Director, Head of Quality Management, Sensirion AG

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